

LOOPS

PRINT FROM 1 TO 5

```
#include<iostream>
using namespace std;

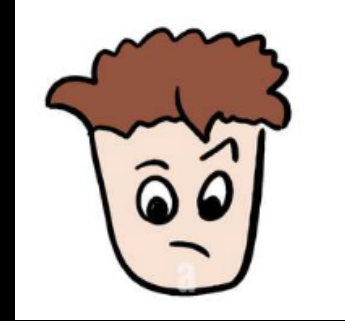
int main()
{
    cout << 1 << endl;
    cout << 2 << endl;
    cout << 3 << endl;
    cout << 4 << endl;
    cout << 5 << endl;
}
```

No double quote
gpl



PRINT FROM 1 TO 10

```
int main()
{
    cout << 1 << endl;
    cout << 2 << endl;
    cout << 3 << endl;
    cout << 4 << endl;
    cout << 5 << endl;
    cout << 6 << endl;
    cout << 7 << endl;
    cout << 8 << endl;
    cout << 9 << endl;
    cout << 10 << endl;
}
```



PRINT FROM 1 TO 1000

```
int main()
{
    cout << 1 << endl;
    cout << 2 << endl;
    cout << 3 << endl;
    cout << 4 << endl;
    cout << 5 << endl;
    cout << 6 << endl;
    cout << 7 << endl;
    cout << 8 << endl;
    cout << 9 << endl;
    cout << 10 << endl;
```

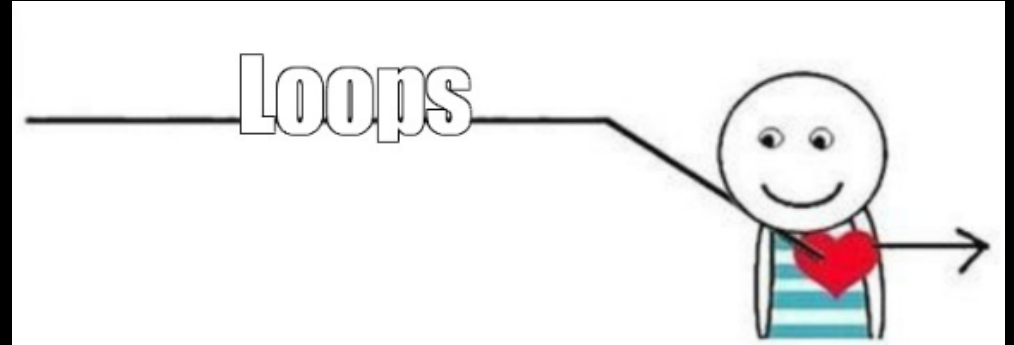
·
·
·
·
·



How can we repeat work without
writing the same code again and
again?

LOOPS

When you want to
repeat an action
again and again
till a certain condition,
we can use loops.



TYPE OF LOOPS

1) **While** loop

2) **For** loop

3) **Do-while** loop

WHILE LOOP

```
initialization;
```

```
while(condition)
```

```
{
```

```
    // body
```

```
    update;
```

```
}
```

PRINT FROM 1 TO 10

```
int i = 1; False
while(i <= 10)
{
    cout << i << endl;
    i++;
}
```

1
2
3
4
5

6
7
8

9
10

11 <= 10 *No*

i

1 2 3 4 5 6 7 8 9 10 11

1 <= 10 YES

2 <= 10 YES

3 <= 10 YES

4 <= 10 YES

5 <= 10 YES

PRINT 1 TO N

```
int main()
{
    int n;
    cin >> n;

    int i = 1;
    while(i <= n)
    {
        cout << i << endl;
        i++;
    }
}
```

]

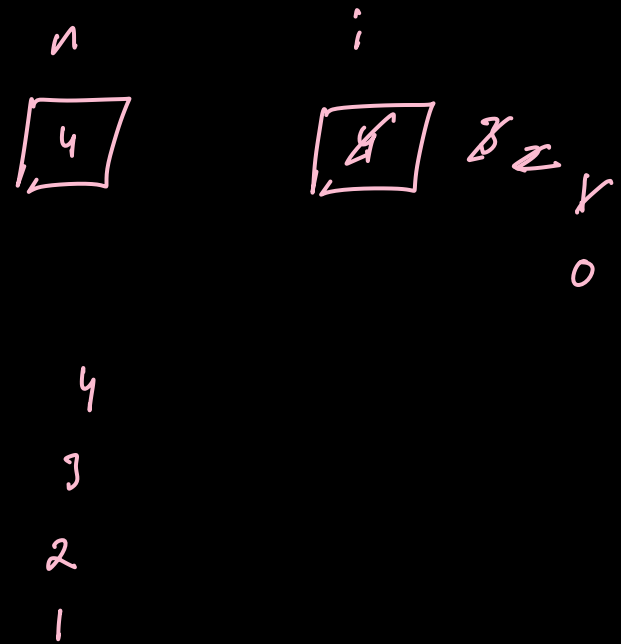


```
int i = n;
while ( i >= 1)
{
    cout << i << endl;
    i--;
}
```

PRINT N TO 1

```
int main()
{
    int n;
    cin >> n;

    int i = n;
    while(i >= 1)
    {
        cout << i << endl;
        i--;
    }
}
```



PRINT NUMBERS FROM L TO R

```
int main()
{
    int l, r;
    cin >> l >> r;

    int i = l;
    while(i <= r)
    {
        cout << i << endl;
        i++;
    }
}
```

PRINT ALL EVEN NUMBERS FROM 1 TO N

n = 5

2

4

n = 6

2

4

6

```
int n;  
cin >> n;  
  
int i = 1;  
while(i <= n)  
{  
    if(i % 2 == 0)  
    {  
        cout << i << endl;  
    }  
    i++;  
}
```

i	n
1	5
2	
3	
4	
5	
6	2
	4

PRINT ALL EVEN NUMBERS FROM 1 TO N

n = 5

2

4

n = 6

2

4

6

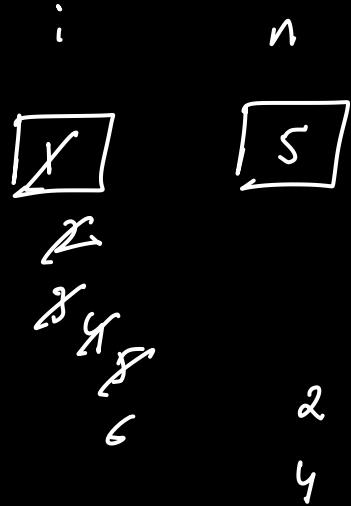
```
int n;  
cin >> n;
```

```
int i = 2;  
while(i <= n)  
{
```

```
    cout << i << endl;
```

```
    i += 2;
```

```
}
```



PRINT ALL UPPERCASE ALPHABETS (A - Z)

```
char ch = 'A';
```

```
while (ch <= 'Z')
```

```
{
```

```
    cout << ch << endl;
```

```
    ch++;
```

```
}
```

ASCII Values

'A' → 65

'a' → 97

'B' → 66

'b' → 98

'C' → 67

'c' → 99

⋮

⋮

PRINT MULTIPLICATION TABLE

n = 5

$$5 \times 1 = 5$$

$$5 \times 2 = 10$$

$$5 \times 3 = 15$$

$$5 \times 4 = 20$$

$$5 \times 5 = 25$$

$$5 \times 6 = 30$$

$$5 \times 7 = 35$$

$$5 \times 8 = 40$$

$$5 \times 9 = 45$$

$$5 \times 10 = 50$$

n = 6

$$6 \times 1 = 6$$

$$6 \times 2 = 12$$

$$6 \times 3 = 18$$

$$6 \times 4 = 24$$

$$6 \times 5 = 30$$

$$6 \times 6 = 36$$

$$6 \times 7 = 42$$

$$6 \times 8 = 48$$

$$6 \times 9 = 54$$

$$6 \times 10 = 60$$

```
int i=1;
```

```
while(i <= 10)
```

```
{
```

```
cout << n << " x " << i << " = " <<
```

```
i++;
```

```
}
```

$(n \times i)$ << endl;

PRINT NUMBER IN REVERSE

$n = 1950$

$n = 512$

215

$n = 1927$

7291

$n = 3429$

9243

```
int n;
```

```
cin >> n;
```

```
while ( n != 0 )
```

```
{
```

```
    cout << n % 10;
```

```
    n = n / 10;
```

```
}
```

lastDigit = $n \% 10$

after division by 10, removes the last

n

1927

192

↓

19

↓

↓

↓

0

DIGITS SUM

Sum

0

$$n = 512 \Rightarrow 5 + 1 + 2 = 8$$

8

$$n = 1927 \Rightarrow 1 + 9 + 2 + 7 = 19$$

19

$$n = 3429 \Rightarrow 3 + 4 + 2 + 9 = 18$$

18

```
int sum = 0;
```

DSA Contd

```
int n;
```

```
cin >> n;
```

28th Jan

Wed

8-10 PM

```
while ( n != 0 )
```

```
{
```

```
    sum += n % 10;
```

```
    n = n / 10;
```

```
}
```

```
cout << sum << endl;
```

REVERSE AND STORE IN A VARIABLE

n = 512

ans = 215

n = 1927

ans = 7291

n = 3429

ans = 9243

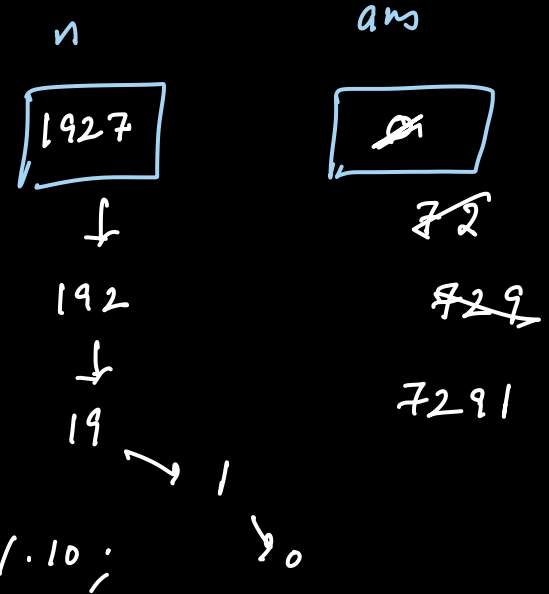
```
int ans = 0;

int n;
cin >> n;

while ( n != 0 )
{
    ans = (ans * 10) + n % 10;

    n = n / 10;
}

cout << ans << endl;
```



PALINDROME

n = 12321

YES

rev = 12321

n = 1927

NO

rev = 7291

n = 1

YES

rev = 1

```
int rev = 0;
```

```
int n;
```

```
cin >> n;
```

```
int temp = n;
```

```
while ( n != 0 )
```

```
{    rev = (rev * 10) + n % 10;
```

```
    n = n / 10;
```

```
}
```

```
if ( rev == n ) → temp
```

```
{    cout << "YES";
```

```
}
```

```
else
```

```
{
```

what will be the
value of N?



FOR LOOP

```
    cout << "No";  
}
```

```
for(initialization; condition; update) {  
    // code  
}
```

int i = 1;

while (i <= 100)

{

cout << i << endl;

i++;

}

i
1 2 3

for (int i = 1; i <= 100; i++)

{

cout << i << endl;

}

while ()  for loop 


PATTERN PRINTING

PRINT 1 TO N

Given n (number of rows), print the following pattern:

$n = 5$

1

2

3

4

5

```
for(i = 1; i <= n; i++)  
{  
    cout << i << endl;  
}
```

$n = 6$

1

2

3

4

5

6

$n = 7$

1

2

3

4

5

6

7

2 STARS

Given n (number of rows), print the following pattern:

$n = 5$

**

**

**

**

**

```
For(int i=1; i<=n ; i++)  
{  
    cout << "**" << endl;  
}
```

$n = 6$

**

**

**

**

**

**

$n = 7$

**

**

**

**

**

**

**

4 STARS

Given n (number of rows), print the following pattern:

$n = 5$

$n = 6$

$n = 7$

M STARS (RECTANGLE)

Given **n** (no. of rows) and **m** (no. of cols) , print the following pattern:

n = 5, m = 7

```
*****  
*****  
*****  
*****  
*****
```

n = 6, m = 3

```
***  
***  
***  
***  
***  
***
```

n = 7, m = 4

```
****  
****  
****  
****  
****  
****  
****
```

SQUARE

Given n (no. of rows and cols), print the following pattern:

$n = 5$

```
* * * * *  
* * * * *  
* * * * *  
* * * * *  
* * * * *
```

$n = 6$

```
* * * * *  
* * * * *  
* * * * *  
* * * * *  
* * * * *  
* * * * *
```

$n = 7$

```
* * * * *  
* * * * *  
* * * * *  
* * * * *  
* * * * *  
* * * * *  
* * * * *
```

PYRAMID

Given n (no. of rows), print the following pattern:

$n = 5$

```
*  
**  
***  
****  
*****
```

$n = 6$

```
*  
**  
***  
****  
*****  
*****
```

$n = 7$

```
*  
**  
***  
****  
*****  
*****  
*****
```

INVERTED PYRAMID

Given n (no. of rows), print the following pattern:

$n = 5$

**

*

$n = 6$

**

*

$n = 7$

**

*

HOLLOW SQUARE

Boundary will exist

Given n (no. of rows and cols), print the following pattern:

$n = 5$

$i=1$ $j=1$

```
* * * * *
```

$j=4$

```
* * * * *
```

$n = 6$

```
* * * * * *
```

```
* * * * *
```

```
* * * * *
```

```
* * * * *
```

```
* * * * *
```

```
* * * * *
```

HOLLOW RECTANGLE

Given n (no. of rows) and m (no. of cols) , print the following pattern:

$n = 5, m = 7$

```
*****  
*       *  
*       *  
*       *  
*****
```

$n = 6, m = 8$

```
*****  
*       *  
*       *  
*       *  
*       *  
*****
```

$n = 7, m = 4$

```
****  
*  *  
*  *  
*  *  
*  *  
*  *  
****
```

NUMBERED RECTANGLE

Given n (no. of rows) and m (no. of cols) , print the following pattern:

$n = 5, m = 7$

1111111	x x x x x x x
2222222	x x x x x x x
3333333	x x x x x x x
4444444	x x x x x x x
5555555	x x x x x x x

$n = 6, m = 3$

111
222
333
444
555
666

$n = 7, m = 4$

1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
5 5 5 5
6 6 6 6
7 7 7 7

- 1) Draw the star pattern
- 2) Replace $x \rightarrow$ given elem

NUMBERED RECTANGLE - II

Given n (no. of rows) and m (no. of cols) , print the following pattern:

$n = 5, m = 7$

1 2 3 4 5 6 7

1 2 3 4 5 6 7

1 2 3 4 5 6 7

1 2 3 4 5 6 7

1 2 3 4 5 6 7

$n = 6, m = 3$

1 2 3

1 2 3

1 2 3

1 2 3

1 2 3

1 2 3

$n = 7, m = 4$

1 2 3 4

1 2 3 4

1 2 3 4

1 2 3 4

1 2 3 4

1 2 3 4

1 2 3 4

NUMBERED RECTANGLE - III

Given n (no. of rows) and m (no. of cols) , print the following pattern:

$n = 5, m = 7$

A B C D E F G

A B C D E F G

A B C D E F G

A B C D E F G

A B C D E F G

$n = 6, m = 3$

A B C

A B C

A B C

A B C

A B C

A B C

$n = 7, m = 4$

A B C D

A B C D

A B C D

A B C D

A B C D

A B C D

A B C D

NUMBERED RECTANGLE – IV

Given n (no. of rows) and m (no. of cols) , print the following pattern:

$n = 5, m = 7$

```
AAAAAAA  
BBBBBBB  
CCCCCCC  
DDDDDDD  
EEEEEEE
```

$n = 6, m = 3$

```
AAA  
BBB  
CCC  
DDD  
EEE  
FFF
```

$n = 7, m = 4$

```
AAAA  
BBBB  
CCCC  
DDDD  
EEEE  
FFFF  
GGGG
```

NUMBERED PYRAMID

Given **n** (no. of rows), print the following pattern:

n = 5

1

22

333

4444

55555

n = 6

1

22

333

4444

55555

666666

n = 7

1

22

333

4444

55555

666666

7777777

BINARY PYRAMID

Given n (no. of rows), print the following pattern:

$n = 5$

```
0
10
010
1010
01010
```

$n = 6$

```
0
10
010
1010
01010
101010
```

$n = 7$

```
0
10
010
1010
01010
101010
0101010
```